three paper: Antras

2026 年 3 月 13 日

Introduction

three paper

1. ”Measuring the Average Period of Production” (joint with Vitalii Tubdenov), American Economic Association Papers and Proceedings, Vol. 115, May 2025, pp. 624–30).

2. ”Interest Rates and World Trade: An ‘Austrian’ Perspective” American Economic Association Papers and Proceedings, Vol. 113, May 2023, pp. 65-69.

3. ”An Austrian’ Model of Global Value Chains” January 2023, reject and resubmit at the American Economic Review.

Austrian

Austrian!

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Janek Wasserman’s “The Marginal Revolutionaries: How Austrian Economists Fought the War of Ideas” which inspired Antras to write this paper.

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Production takes time.

then, what research

Interest Rates and World Trade: An 'Austrian' Perspective*

Pol Antràs
Harvard University and NBER

January 11, 2023

Abstract

An 'Austrian' Model of Global Value Chains*

Pol Antràs
Harvard University and NBER

January 26, 2023

Abstract

Attachments

Austrian_March_10_2026.pdf

Austrian_Jan_26_2023.pdf

Austrian_2025.p

another one

Measuring the Average Period of Production*

Pol Antràs
Harvard University and NBER

Vitalii Tubdenov
Harvard University

February 2025

Measuring the Average Period of Production

Prodcution process

a production take place in time interval[0,T]

T :length of the prodcution process

the mapping between inputs and outputs, production function:

$$Y(\boldsymbol{L}) = Z(\boldsymbol{L})F(\boldsymbol{L})$$

where,

$$\boldsymbol{L} = \{\ell(t)\}_{t\in[0,T]}$$

$F(\boldsymbol{L})$: producution technology, homogeneous of degree one in $\boldsymbol{L}$
$Z(\boldsymbol{L})$: productivity, homogeneous of degree 0 in $\boldsymbol{L}$

Production process

define $\boldsymbol{\lambda} = \{\ell(t)/L\}_{t\in[0,T]}$,

where $L = \int_0^T \ell(t)\, dt$, total labor use along the whole production process.

Thus, $\boldsymbol{\lambda}(t)$captures the distribution of labor expenditures along the production process.

the measure!

$$\mathcal{APP}(\boldsymbol{\lambda}) = \int_0^T (T - t)\lambda(t)\, dt.$$

represents a weighted average temporal distance

Relationship between APP and interest rates

in the paper, long process

Empirical Measure of the APP

then, develop an empirical counterpart to our measure of the average period of production

$$C(\hat{t}) = \int_0^{\hat{t}} w\ell(t)\, dt.$$

cumulative cost of the good in interval $[0, \hat{t}]$

Total Inventries

- at each instant t
- the firm begins the production of N goods
- continues to add value to goods begun in previous periods $t\prime \in (t-T, t)$
- and complete the production of N goods that begun at $t-T$

$$I = N \int_0^T C(\hat{t})\, d\hat{t} = Nw \int_0^T \int_0^{\hat{t}} \ell(s)\, ds\, d\hat{t},$$

COGS

at each instant t

$$C(T) = NwL$$

corresponds to the accounting concept of the cost of goods sold (or COGS)

the ratio of inventories to the cost of goods sold

the ratio of inventories to the cost of goods sold

$$\frac{I}{C(T)} = \int_0^T \int_0^{\hat{t}} \frac{\ell(s)}{L}\, ds\, d\hat{t} = \int_0^T \int_0^{\hat{t}} \lambda(s)\, ds\, d\hat{t}.$$

Solving this double integral by changing the order of integration yields

$$\frac{I}{C(T)} = \int_0^T \int_0^{\hat{t}} \lambda(s)\, ds\, d\hat{t} = \int_0^T \int_s^T \lambda(s)\, d\hat{t}\, ds = \int_0^T \lambda(s)(T-s)\, ds = \mathcal{APP}.$$

GET! Empirical measure of the APP

The Average Period of Production in the U.S.

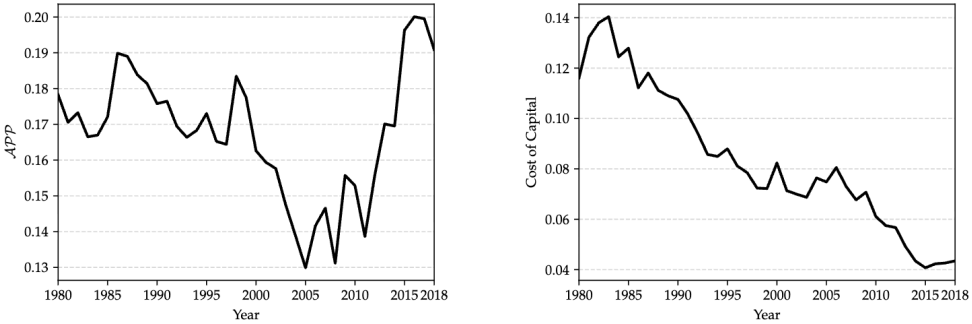
Data Source

using annual financial reports from publicly traded U.S. firms for the period 1980–2018. This data is obtained from the Compustat North America database (SP Global, 2025).

we focus on presenting results based on the sample of all firms in Compustat classified as belonging to the manufacturing sector (NAICS codes 31–32–33).

Aggregte trends

Figure 1: $\mathcal{APP}$ and Cost of Capital in U.S Manufacturing

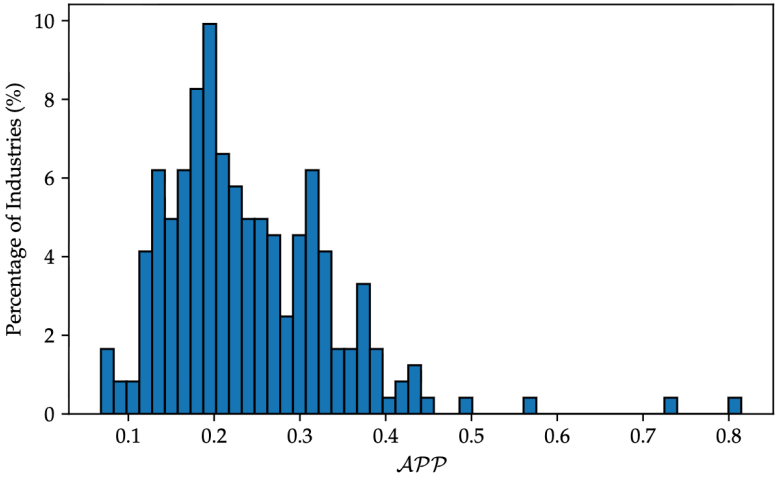


(a) Cost-Weighted Average Period of Production **(b)** Cost-Weighted Average Cost of Capital

Notes: The figure shows the cost-weighted annual averages of $\mathcal{APP}$ and cost of capital for U.S. publicly traded companies in manufacturing industries from 1980 to 2018. $\mathcal{APP}$ represents the ratio of inventories to COGS, while cost of capital is the ratio of interest expenses to total debt (short- and long-term). Source: Compustat.

Variation in the Average Period of Production Across Industries

Figure 2: Distribution of $\mathcal{APP}$ in U.S. Manufacturing



Notes: The figure shows the distribution of industries ($N = 242$) based on $\mathcal{APP}$. For each industry, defined as a six-digit NAICS code, $\mathcal{APP}$ is calculated as a cost-weighted average across firms within each year, then

Variation in the Average Period of Production Across Industries

Table 1: The Top 10 U.S. Manufacturing Industries by $\mathcal{APP}$

NAICS	Industry	$\mathcal{APP}$
312140	Distilleries	0.814
312130	Wineries	0.737
339115	Ophthalmic Goods Manuf.	0.573
312230	Tobacco Manuf.	0.488
333997	Scale and Balance Manuf.	0.441
333132	Oil and Gas Field Machinery and Equipment Manuf.	0.439
316998	All Other Leather Good and Allied Product Manuf.	0.432
325412	Pharmaceutical Preparation Manuf.	0.430
334510	Electromedical and Electrotherapeutic Apparatus Manuf.	0.421
332215	Metal Kitchen Cookware, Utensil, Cutlery, and Flatware Manuf.	0.415

Notes: The table lists the top 10 U.S. manufacturing industries ranked by $\mathcal{APP}$. For each industry, defined as a six-digit NAICS code, $\mathcal{APP}$ is calculated as a cost-weighted average across firms within each year, then averaged from 1980 to 2018. Industries with fewer than 50 firm-year observations are excluded. Please refer to the replication package for the complete ranking. Source: Compustat.

Variation in the Average Period of Production Across Industries

Table 2: The Bottom 10 U.S. Manufacturing Industries by $\mathcal{APP}$

NAICS	Industry	$\mathcal{APP}$
324110	Petroleum Refineries	0.068
311511	Fluid Milk Manuf.	0.069
311812	Commercial Bakeries	0.071
325193	Ethyl Alcohol Manuf.	0.073
336370	Motor Vehicle Metal Stamping	0.083
312112	Bottled Water Manuf.	0.097
327992	Ground or Treated Mineral and Earth Manuf.	0.112
336350	Motor Vehicle Transmission and Power Train Parts Manuf.	0.112
337214	Office Furniture (except Wood) Manuf.	0.115
311520	Ice Cream and Frozen Dessert Manuf.	0.116

Notes: The table lists the bottom 10 U.S. manufacturing industries ranked by $\mathcal{APP}$. For each industry, defined as a six-digit NAICS code, $\mathcal{APP}$ is calculated as a cost-weighted average across firms within each year, then averaged from 1980 to 2018. Industries with fewer than 50 firm-year observations are excluded. Please refer to the replication package for the complete ranking. Source: Compustat.

The Average Period of Production and the Cost of Capital

$$\mathcal{APP}_{it} = \beta R_{it} + \gamma \mathbf{Z}_{it} + \mu_i + \lambda_t + \varepsilon_{it},$$

R_{it} is the cost of capital faced by firm i in year t

The Average Period of Production and the Cost of Capital

Table 4: Summary Statistics for Firms in U.S. Manufacturing, 1980–2018

	Mean	Std. Dev.	P25	Median	P75	N
Average Period of Production	0.310	0.243	0.161	0.251	0.388	57,800
Cost of Capital	0.106	0.082	0.064	0.090	0.121	57,800
IT Capital Intensity	0.048	0.049	0.014	0.022	0.072	46,135
Capital Intensity	166.4	257.9	49.15	85.58	163.5	56,273
Skill Intensity	0.362	0.140	0.242	0.356	0.470	57,791
Labor Productivity	193.9	272.0	81.03	121.5	203.9	57,791

The Average Period of Production and the Cost of Capital

Table 5: $\mathcal{APP}$ and Cost of Capital in U.S. Manufacturing

Dependent Variable:	Average Period of Production		
	(1)	(2)	(3)
Cost of Capital	-0.090^a	-0.088^a	-0.090^a
	(0.007)	(0.007)	(0.007)
IT Capital Intensity	–	0.006	0.005
	–	(0.020)	(0.019)
Capital Intensity	–	–	−0.030
	–	–	(0.028)
Skill Intensity	–	–	−0.046
	–	–	(0.049)
Labor Productivity	–	–	−0.045
	–	–	(0.047)
R^2	0.789	0.810	0.812
Observations	57,800	46,135	44,608

Notes: The table presents the results of estimating equation (19) on a panel of U.S. publicly traded manufacturing companies from 1980 to 2018. All variables are in logarithms. All regressions include firm and year fixed effects. The standard errors are clustered at the industry (six-digit NAICS codes) and year levels (two-way clustering).

The Average Period of Production and the Cost of Capital

Table 6: $\mathcal{APP}$ and Cost of Capital by Inventory Type in U.S. Manufacturing

	Total Inventories to COGS (1)	Raw Materials to COGS (2)	Work-in-Process to COGS (3)	Finished Goods to COGS (4)
Cost of Capital	-0.101^a	-0.100^a	-0.123^a	-0.107^a
	(0.006)	(0.009)	(0.011)	(0.010)
R^2	0.793	0.766	0.796	0.763
Observations	37,418	37,418	37,418	37,418

Notes: The table presents the results of estimating equation (19) on a panel of U.S. publicly traded companies in manufacturing from 1980 to 2018. The dependent variables are the ratios of certain types of inventory to COGS. We estimate each regression on the subset of observations that contain complete data on inventory types. All variables are in logarithms. All regressions include firm and year fixed effects. The standard errors are clustered at the industry (six-digit NAICS codes) and year levels (two-way clustering). Statistical significance at the 1%, 5%, and 10% levels is denoted by superscripts a, b, and c, respectively. Source: Compustat.

The Average Period of Production Worldwide

Rank Correlations of Industries

Table 7: Rank Correlations of Manufacturing Industries by $\mathcal{APP}$ Across Countries

	CAN	CHN	DEU	FRA	GBR	IND	JPN	KOR	MYS	TWN	USA
CAN	–	0.41	–	–	0.57^c	0.64^b	0.75^a	–	–	0.57^c	0.71^a
CHN	0.41	–	–	–	0.42	0.42^b	0.50^a	0.69^a	–	0.47^b	0.64^a
DEU	–	–	–	–	–	–	–	–	–	–	0.64^b
FRA	–	–	–	–	–	–	–	–	–	–	0.87^a
GBR	0.57^c	0.42	–	–	–	0.83^a	0.62^a	0.71^b	–	0.41	0.76^a
IND	0.64^b	0.42^b	–	–	0.83^a	–	0.43^a	0.45^b	0.38	0.29	0.42^a
JPN	0.75^a	0.50^a	–	–	0.62^a	0.43^a	–	0.73^a	0.21	0.34^b	0.57^a
KOR	–	0.69^a	–	–	0.71^b	0.45^b	0.73^a	–	–	0.58^a	0.69^a
MYS	–	–	–	–	–	0.38	0.21	–	–	–	0.05
TWN	0.57^c	0.47^b	–	–	0.41	0.29	0.34^b	0.58^a	–	–	0.61^a
USA	0.71^a	0.64^a	0.64^b	0.87^a	0.76^a	0.42^a	0.57^a	0.69^a	0.05	0.61^a	–

Notes: Spearman rank correlations are reported. To construct the rankings of industries (defined as six-digit NAICS codes) for each country, we first calculate the cost-weighted average of $\mathcal{APP}$ for each country-industry pair within each year from 1980 to 2018 and then compute the average of these values across those years. We limit the analysis to country-industry pairs with at least 50 observations during this period. Rank correlations are reported only for country pairs with at least 10 overlapping industries. The analysis is based on a panel of publicly traded manufacturing companies. Statistical significance at the 1%, 5%, and 10% levels is denoted by superscripts a, b, and c, respectively. Source: Compustat.

Pooled Regressions

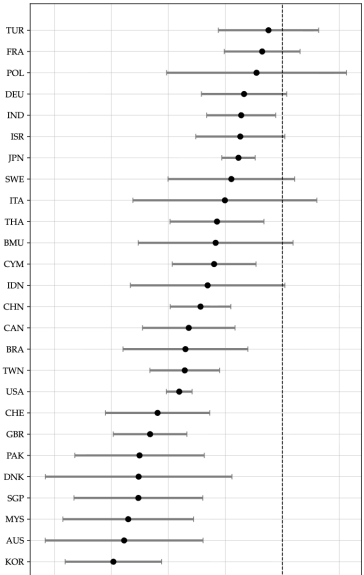
Table 8: $\mathcal{APP}$ and Cost of Capital Worldwide

Dependent Variable:	Average Period of Production		
	(1) U.S. only	(2) All Countries	(3) U.S. excluded
Cost of Capital	-0.090^a	-0.079^a	-0.071^a
	(0.007)	(0.005)	(0.005)
R^2	0.789	0.786	0.786
Observations	57,800	180,954	123,154

Notes: The table presents the results of estimating equation (19) on a panel of global publicly traded companies in manufacturing from 1980 to 2018. The first column uses data for U.S. firms, the second includes firms from all countries, and the last excludes U.S. firms. All variables are in logarithms. All regressions include firm and year fixed effects. The standard errors are clustered at the industry (six-digit NAICS codes) and year levels (two-way clustering). Statistical significance at the 1%, 5%, and 10% levels is denoted by superscripts a, b, and c, respectively. Source: Compustat.

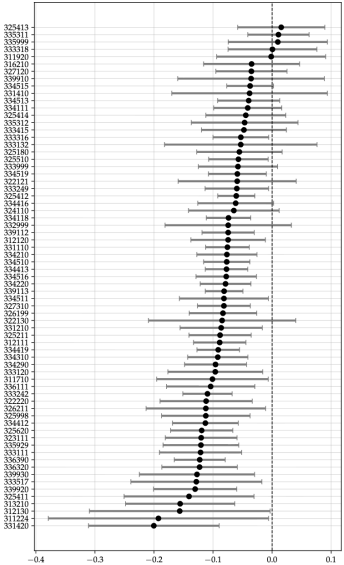
Heterogeneity at the Industry and Country Levels

Figure 4: $\mathcal{APP}$ and Cost of Capital in Manufacturing: Country Heterogeneity



Heterogeneity at the Industry and Country Levels

Figure 5: $\mathcal{APP}$ and Cost of Capital Worldwide: Industry Heterogeneity



Conclusion

Our ultimate aim is to foster further empirical research into how production length and interest rates shape industrial structure.

(old Version) An ‘Austrian’ Model of Global Value Chains

Baseline model

Building blocks

- a sequential production process(Antràs and de Gortari (2020))
- a temporal dimension to production (Findlay (1978))
- to close the model, a supply side to the capital market(Antràs and Caballero (2009,2010))

Environment

the population mass is constant and equal to L

All agents are endowed with one unit of labor

Consumers value a single final good, which is taken to be the numéraire.

Multi-stage production

production of the single final good needs to undergo N stages in a pre-determined, sequential order

At each stage $n > 1$, production combines labor with the good finished up to the previous stage $n - 1$.

production in the initial stage n 1 only uses labor

Multi-stage production

$$y_n = (z_n L_n)^{\alpha_n} (y_{n-1})^{1-\alpha_n},$$

where y_n is output after stage n is performed.

z_n denotes labor productivity at stage n

Production takes time

The main innovationis in acknowledging that production takes time.

the more time is devoted to production, the more output will be obtained

following the approach in Findlay (1978).

Optimal production length

in stage 1

$$\pi_1 = p_1 z_1(t_1) L_1 e^{-rt_1} - wL_1.$$

Firms treat good and factor prices as given, so t_1 will be set to maximize profit

$$\frac{z_1'(t_1^*)}{z_1(t_1^*)} = r.$$

condition on $z_n(t)$

How does productivity change over time.

$$z_n'(t) > 0$$

$$\frac{z_n''(t)}{z_n'(t)} < 0 < \frac{z_n'(t)}{z_n(t)},$$

to ensure the existence of a unique optimal. $\frac{z_n'(t_1)}{z_n(t_n)} \to \infty$ when $t_n \to 0$ $\frac{z_n'(t_1)}{z_n(t_n)} \to 0$ when $t_n \to \infty$

Optimal production length

in stage n, producers pay wages and purchase the good finished up to the prior stage $n-1$

$$\pi_n = p_n \left(z_n(t_n) L_n\right)^{\alpha_n} \left(y_{n-1}\right)^{1-\alpha_n} e^{-rt_n} - wL_n - p_{n-1} y_{n-1}.$$

then, the optimal stopping time t_n^* will satisfy

$$\alpha_n \frac{z_n'(t_n^*)}{z_n(t_n^*)} = r.$$

Equilibrium Allocation of Labor Across Stages

Having solved all optimal production lengths.

next solve for the allocation of labor across stages

$$wL_n e^{rt_n} = \alpha_n p_n y_n$$
$$p_{n-1} y_{n-1} e^{rt_n} = (1 - \alpha_n) p_n y_n.$$

Equilibrium Allocation of Labor Across Stages

Iterating the second equation forward and plugging the result into the first one, delivers

$$wL_n e^{r \sum_{m=n}^{N} t_m} = \alpha_n \beta_n y_N,$$

where $\beta_n \equiv \prod_{m=n+1}^{N} (1 - \alpha_m)$,, y_N is final output and final good revenue($p_N = 1$)

Equilibrium Allocation of Labor Across Stages

according to production function:

$$y_N = \prod_{n=1}^{N} \left(z_n(t_n) L_n\right)^{\alpha_n \beta_n} .$$

So, for any two stage $n' > n$:

$$\frac{L_{n'}}{L_n} = \frac{\alpha_{n'} \beta_{n'}}{\alpha_n \beta_n} e^{r \sum_{m=n}^{n'-1} t_m} .$$

SO GET! Equilibrium allocation of labor across stages

then

1. finish the closed economy version
2. from close to open
3. trade cost
4. trade credit

物流服务标准化与企业供应链地理布局

- 不同地区物流标准的混乱，带来高昂的物流成本, 也制约了远距离企业之间的合作

- 2014 年逐步试点的物流标准化政策，以推进以托盘共用体系和建设物流综合信息服务平台为切入点，推进物流标准化工作（商务部和国家标准管理委员会，2014）。

- 借助中国 2014 年逐步试点的物流标准化政策，研究物流体系标准化对企业供应链地理布局的影响。
- 机制：内，企业自身的经营成本，信息获取能力；外，优化地区的市场环境，市场一体化程度。
- 进一步，企业的供应链风险，效率等；是否有供应链溢出。

物流服务标准化政策试点	
年份	**试点地区**
2014	北京、上海、广州
2015	天津、石家庄、唐山、杭州、南京、徐州、芜湖、东莞、肇庆、佛山、中山
2016	邯郸、承德、德州、临沂、淄博、无锡、合肥、马鞍山、武汉、黄石、怀化、常德、贵阳、遵义、南昌、九江、厦门、三明

为检验物流标准化政策对企业供应链地理地理分布的影响，设定如下交错型双重差分模型进行识别：

$$Disw_S_{i,j,c,t} = \alpha + \beta Standard_{c,t} + \Gamma X_{i,t} + \delta_i + \lambda_t + \omega_{j,t} + \varepsilon_{i,j,c,t}$$

$$Disw_C_{i,j,c,t} = \alpha + \beta Standard_{c,t} + \Gamma X_{i,t} + \delta_i + \lambda_t + \omega_{j,t} + \varepsilon_{i,j,c,t}$$

核心解释变量：物流标准化，以 2014 年在城市层面分批实施的物流标准化试点政策作为准实验, 构建核心解释变量:。当城市 c 在年份 t 被纳入物流标准化试点时, $Standard_{c,t}$ 在 t 及 t 年之后都赋值为 1, 否则为 0。
被解释变量：企业的供应链地理距离，以企业与前五大供应商和前五大客户的加权距离作为企业供应链地理距离的测度指标。

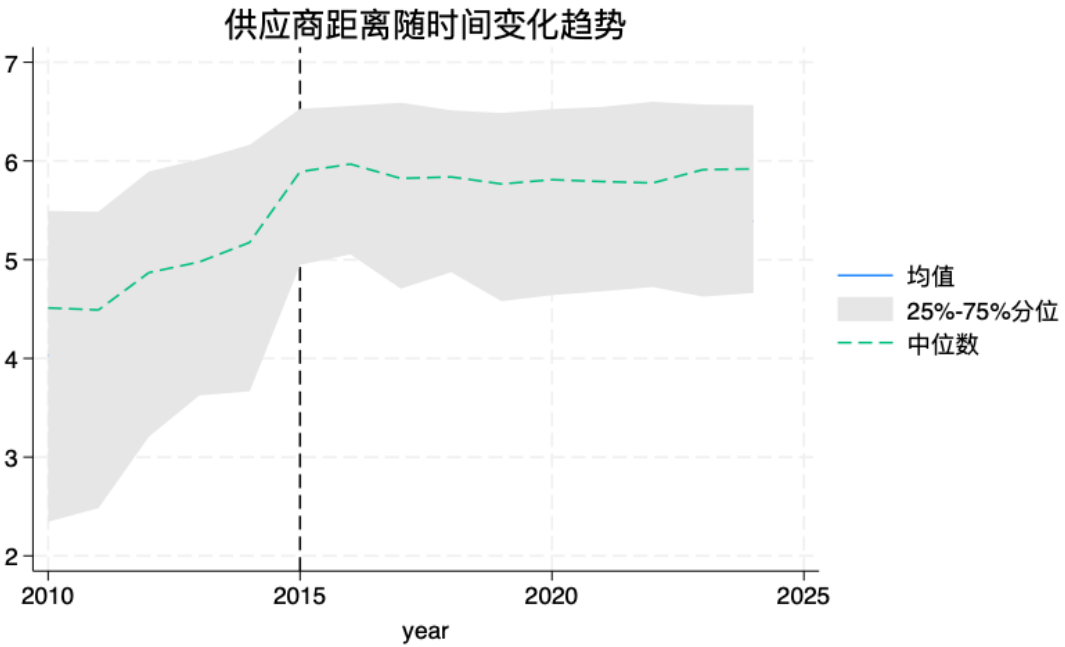
$$Disw_S_{it} = \ln\left(1 + \sum_p Dis_S_{ipt} \times Ratio_S_{ipt}\right)$$

$$Disw_C_{it} = \ln\left(1 + \sum_p Dis_S_{ipt} \times Ratio_S_{ipt}\right)$$

来源国泰安 CSMAR 数据库收集整理的 2010-2024 年 A 股上市公司数据作为研究样本。

表1 描述性统计

变量	样本量	均值	1/4分位数	中位数	3/4分位数	标准差
Disw_s	7369	5.229	4.456	5.682	6.464	1.683
Disw_c	8187	5.026	4.115	5.541	6.403	1.853
Standard	10017	0.241	0.000	0.000	0.000	0.428
Fir	10016	0.221	0.085	0.181	0.322	0.171
Lev	10017	0.415	0.241	0.402	0.572	0.213
Gro	10001	0.150	-0.037	0.096	0.251	0.377
Cash	10016	0.045	0.005	0.044	0.085	0.072
Roa	10017	0.038	0.013	0.038	0.067	0.061
Size	10016	21.887	20.905	21.750	22.731	1.413
Top1	9471	34.851	22.986	32.110	45.341	15.160
Board	9470	9.368	8.000	9.000	10.000	2.282
Indep	9310	0.370	0.333	0.333	0.417	0.050


供应商距离随时间变化趋势
7
6
5
4
3
2
2010
2015
2020
2025
year
均值
25%-75%分位
中位数

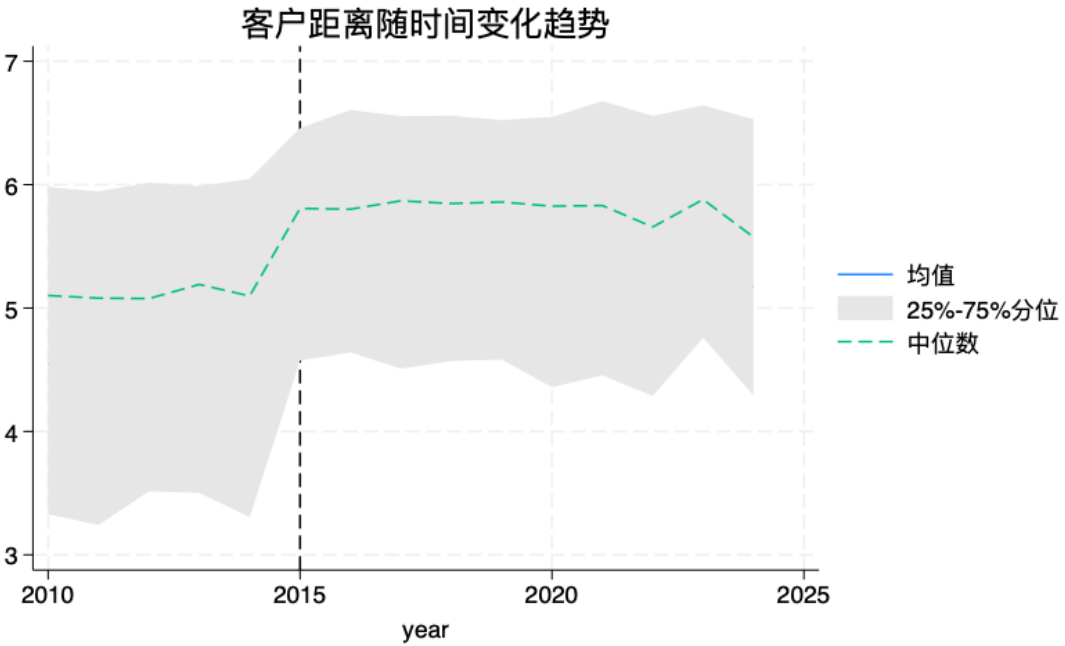

客户距离随时间变化趋势
7
6
5
4
3
2010
2015
2020
2025
year
均值
25%-75%分位
中位数

表 2：基准结果

	(1)	(2)	(3)	(4)	(5)	(6)
	Disw_s	Disw_c	Disw_s	Disw_c	Disw_s	Disw_c
Standard	-0.147*	-0.018	-0.145*	-0.020	-0.150	-0.043
	(0.087)	(0.075)	(0.086)	(0.077)	(0.096)	(0.084)
控制变量	否	否	是	是	是	是
企业固定效应	是	是	是	是	是	是
年份固定效应	是	是	是	是	是	是
行业×年份固定效应	否	否	否	否	是	是
样本量	6835	7639	6256	7076	6127	6912
调整 R^2 值	0.581	0.654	0.589	0.658	0.590	0.666

- 数据缺失问题
- 一种替代效应
- 地方标准的市场分割